

Problema

Pentru o aplicatie de tip elearning trebuie implementat un sistem de gestiune. Sistemul va fi folosit de studenti, profesori si administratori, care trebuie sa introduca un user si o parola pt acces. Administratorul va gestiona conturile studentilor si profesorilor si va adauga cursuri noi, la care va asigna un profesor si o lista de studenti. Un curs se compune dintr-o serie de documente suport si un examen cu mai multe intrebari, la care studentii trebuie sa scrie un raspuns in format text.

Studentii pot participa la unul sau mai multe cursuri, pot sa vizioneze materialele de curs si pot trimite temele spre evaluare.

Un profesor poate sa editeze continutul cursului incarcand documentele si poate corecta temele studentilor inscrisi la curs.

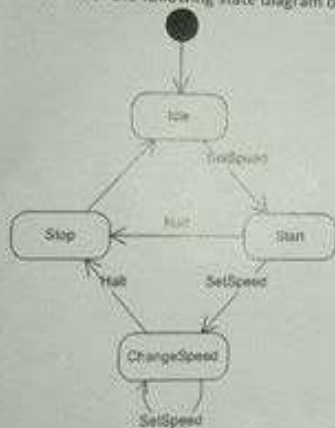
Cand profesorul termina de evaluat o tema studentul va fi notificat real time cu un mesaj.

1. (2 puncte) Sa se identifice actorii si sa se realizeze diagrama cazurilor de utilizare pentru sistem. Sa se realizeze o descriere a cazului de utilizare: „Profesorul corecteaza tema unui student”
2. (3 puncte) Sa se realizeze arhitectura conceputa a sistemului si diagrama de clase, tinand cont de urmatoarele cerinte:
 - Se va utiliza o arhitectura client-server
 - Se va folosi un pattern de access la baza de date (de exemplu Table date Gateway sau Row data gateway).
 - Se va utiliza un design pattern pentru a notifica in timp real studentii cand un administrator posteaza un curs.
3. (1 punct) Realizati diagrama bazei de date pentru aplicatia ceruta
4. (1 punct) Realizati diagrama de deployment pentru aplicatia ceruta
5. (2 puncte) Cod (utilizati orice limbaj de programare doriti):

Sa se evidentieze in cod folosirea design pattern-ului ales mai sus pentru notificarea studentilor atunci cand profesorul termina de corectat o tema.

Software Design 2017

1. Web customers want to use a platform to search for an item with a textual description, and to buy an item from the e-commerce web service that gives the best price. As web services, the eBay auction house and Amazon are available. Analyze possible architectural alternatives and propose a solution. Sketch a diagram to describe the architecture of your system.
2. Consider the following state diagram of a motor:



Design a class model that captures this behavior. Briefly explain what design patterns you used. Show in your model who's responsible for state transitions. Make sure that you have just one object for each state that is reused whenever the context gets into that state.

3. Discuss possible extension points of the Abstract factory DP (what extensions are easily accommodated, what are not).
4. Consider the following code. What class design principle is not respected? Provide a corrected code solution.

```

public interface IDuck {
    void swim();
}

//The calling code
void MakeDuckSwim(IDuck duck) {
    duck.swim();
}

public class Duck implements IDuck {
    public void swim() {
        //do something to swim
    }
}

public class ElectricDuck implements IDuck {
    public void swim() {
        //do something to swim
    }
}

//The corrected code
void MakeDuckSwim(IDuck duck) {
    duck.swim();
}
    
```

Software Design 2017

Part I (if you did not take the Midterm, solve this part too)

1. In a web-based car rental management system, the client can browse a list of available car types providing a period of time and location, can select a car type and see its details and can reserve a selected car. The client can also cancel its reservation. Based on a layered architecture sketch a design solution covering all the 3 layers. Discuss at least 2 advantages and 2 disadvantages of your solution.
2. Given the database schema in fig below, give at least 2 ways to design the Business Logic Layer. Discuss pros and cons for each.



Part II (if you took the Midterm, solve only this part)

3. You have implemented a very complex component that takes some input, does some calculations and saves the results into a text file. You want to sell the component, but first you want to restrict the access to the results by saving the contents in the result text file using some encryption algorithm. a) What design patterns can be used here to make the component use transparent to the user? Motivate. b) Suppose you want to be able to use different encryption algorithms. What design pattern(s) can be used here to vary the encryption algorithm easily? c) Sketch the UML diagram of the entire system with the design patterns you have chosen in task a) and b).
4. A program includes a Java class that defines a static method implementing a clever sorting algorithm over integer arrays that works especially well when sorting 1,000,000 or more integers. Your boss asks you to modify the program to (a) use this method for 1,000,000+ long integer arrays, (b) use quicksort for integer arrays from size 10 to 1,000,000. Would this change be constructed using design pattern(s), generic(s), or inheritance? Why? Explain based on design principles.

Problem

Design a simplified version of Emag or Amazon - marketplace with shopping-cart. The application can be Web or Mobile. Users can browse/view products, add/remove/modify-quantity of products in the Shopping-Cart. Once the checkout button is pressed, the cart is transformed into an Order and saved to DB. The cart is persisted to the DB until the Order is created. An order is always final. Requirements:

1. Draw the E-R diagram (database schema, with multiplicity and PK/FK)
2. Draw the conceptual architecture for Web or Mobile. Please use 'Layers' and another architectural style (explain which style & why)
3. Draw the deployment diagram and the package diagram. Please highlight/group-by layers
4. Select one of the following sub-model/relationships:
 - Shopping-Cart <-> Product
 - Order <-> Product

And

- a) explain the multiplicity (1-1, 1-*, *-1, *-*) and the type of relationship (association/aggregation/composition)
- b) create the class diagram based on a) (only for this sub-model, the data-object/entities sub-layer).
- c) write the Java/C# code for all the entities involved in this scenario (only attributes, not getter/setters), according to your answers on a) and b).

5. Draw the sequence diagram for the flow add-X, add-Y, remove-Y, modify-quantity-X

Other considerations:

- the design should follow the best practices for database design and OOP modelling (you can add other elements' if needed)
- more explanations can be provided at the beginning of the exam - please ask

1. Consider the following code snippet. Is there any class design principle violated? If so, how would you fix this?

```
public void driveCar(Car car) {  
    if(car instanceof PersonalCar){  
        ((PersonalCar) car).drive();  
    }else if(car instanceof SportsCar){  
        ((SportsCar) car).driveFast();  
    }  
}
```

2. Discuss briefly (max 1/2 page) the similarities and differences between the **State** and **Strategy** design patterns.
3. You need to build software for a digital speedometer for boats. The software should be flexible to support various number and types of displays (from simple LED-displays to large screens), and various ways of graphical designs of showing speed on the display. What architectural pattern would you apply? Justify. Provide a draft architecture.
4. You are implementing a file system. The main abstractions in your design would be files and directories. Directories can contain zero or more files or directories. You want to treat directories and files in a uniform way, e.g., both will have name and will provide operations to stream content in and out. What design pattern could be used to achieve this design? Please explain your design by giving a class diagram.

Problema

HipMenu este cel mai folosit sistem online de comanda de mancare din Cluj Napoca. Examenul vostru consta in proiectarea software a sistemului. Sistemul are doua feluri de utilizatori autentificati:

- Clientul, care poate: vizualiza restaurante, deschide o comanda, participa la o comanda, adauga itemi la comanda, checkout comanda, vizualiza istoric si status comenzi.
- Restaurant admin, care poate: CRUD pe items din meniul, marca o comanda ca livrata, vizualiza rapoare.

HipMenu agrega mai multe restaurante, cu meniul propriu pe categorii. Itemii dintr-o categorie pot fi simpli sau compusi. Itemul are un pret, descriere si imagine atasata. Configurarea unui item compus consta din atasarea la un item de alti itemi din categorii conexe (ex: un preparat de carne se serveste cu o garnitura si o salata). Clientul poate deschide o comanda simpla sau o comanda de grup la un restaurant. In cazul unei comenzi de grup, userii participanti sunt notificati in cazul in care comanda se modifica.

O comanda trece prin cativa pasi: adaugare si configurare item (repetitiv) si checkout, compus din confirmare structura comanda, specificare tip de plata, specificare adresa, specificare detalii comanda, confirmare checkout. Userul poate face undo la un pas o data facut. La comenzi de grup, doar utilizatorul care a initiat comanda poate specifica adresa si face checkout cu conditia ca fiecare din subcomenzi sa fie confirmata ca structura si plata.

Exista doua metode de plata: numerar sau card. In functie de selectia userului, sistemul va procesa plata prin metoda specific aleasa. Dupa confirmarea platii, sistemul trimite un email de confirmare utilizatorilor implicati.

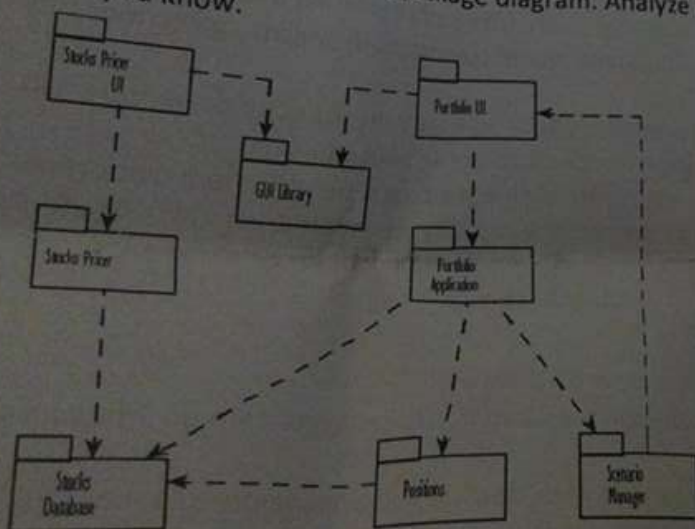
Administratorul poate vedea rapoarte de vanzari pe care le poate exporta in diverse formate.

1. (1p) Identificati actorii si realizati diagrama cazurilor de utilizare. Descrieti un caz de utilizare.
2. (3p) Realizati arhitectura conceptuala a sistemului si diagrama de clase, utilizand:
 - Un pattern architectural studiat la curs, exceptand Layers
 - Doua design patterns specificand descrierea problemei pe care incearca sa o rezolve folosirea lor.
3. (1p) Realizati diagrama de secventa pentru cazul de utilizare "Configurare si adaugare item la shop cart"
4. (1p) Realizati o diagrama a bazei de date pentru aplicatia ceruta
5. (3p) Cod (utilizati orice limbaj de programare doriti):
 - a. (1.5 puncte) Scrieti partea de cod corespunzatoare folosirii unui design pattern de la pct. 2.
 - b. (1.5 puncte) Scrieti partea de cod corespunzatoare cazului de utilizare "Configurare si adaugare item la shop cart" (cuprinzand interactiunea cu baza de date, in orice modalitate studziata).

Software Design 2016 5

Honest team
Alexandra

1. Consider a fractal representation (i.e. an infinite composition). Provide a design including the appropriate design pattern. Decide where to put the composition responsibilities (Add/remove/iterate). Justify.
2. Consider the following package diagram. Analyze its quality considering the package design principles you know.



3. Discuss possible extension points of the Abstract factory DP (what extensions are easily accommodated, what are not)
4. You have to design a system that allows Remote Pair Programming (i.e. two remote programmers work on the same code in real-time: the "driver" edits the code while the "observer" reviews the code and points to objects on the "drivers" desktop.) Roles can be swapped. The work should be saved and the system should ensure that both work on the

same unique copy. Propose a high level architecture for the system including the patterns you consider appropriate. Briefly justify your solution showing where the unique working version should be kept.

Software Design 2016 2

1. Consider a home security system having sensors for: movement, temperature, video camera etc. The monitoring system performs different checks on the sensor data. If a problem is noticed, the system sends a warning to the owner via SMS, or in a dangerous situation may react automatically (e.g. by starting the water sprinkler). Name one architectural pattern that you will use (not design pattern). Draw a diagram that describes your system architecture. Quickly explain in words how the system works (max ¼ pag.). State the two most important advantages of using this architecture. State the two most important disadvantages of using this architecture.
2. Say we have a list of Task entities, and a ProjectTask sub type. Tasks can be closed at any time, except ProjectTasks which cannot be closed once they have a status of Started. Some safeguards are present in the domain:

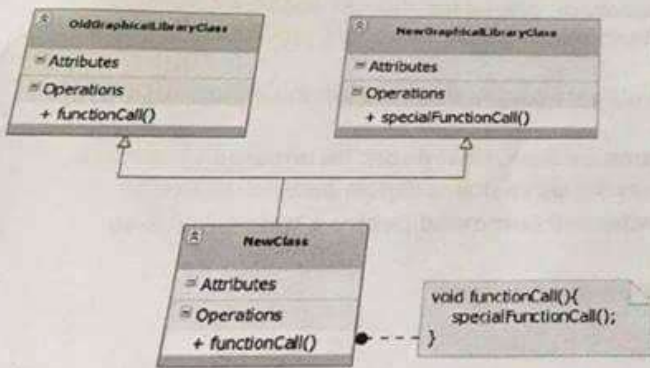
```

public class Task {
    public Status Status { get; set; }
    public virtual void Close()
    {
        Status = Status.Closed;
    }
}

public class ProjectTask : Task {
    public override void Close()
    {
        if (Status == Status.Started)
            throw new Exception("Cannot close a started Project Task");
        base.Close();
    }
}

```

Now when calling Close() on a Task, there is a chance the call will fail if it is a ProjectTask with the started status, when it wouldn't if it was a base Task. What class design principle is violated? How would you fix this?



3. You need to implement and integrate a third-party graphical library into an existing Java Application. Previously the app has used an in-house developed library. Your company hired a highly skilled architect that has given you the following UML diagram. Which design pattern, the architect is trying to suggest you? How would you use the solution proposed by the architect? Will it work?
4. Which pattern has the similar behavior as exceptions handling mechanism?(i.e. The method where the exception occurs could treat the exception. The method where the exception occurs could rethrow it to allow handle it at another layer.) Briefly explain.

Problem

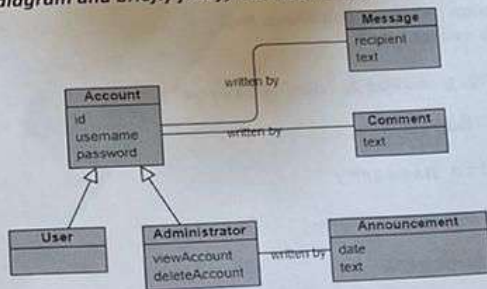
Figlo is an online platform that stores financial data of customers and is able to make financial predictions, based on the data of every client and the financial market. The financial data is represented in the platform as financial cards. Certain types of cards can have linked to them multiple other financial cards. A description of the supported financial cards is following: A card contains, at least, as minimum information, the following attributes: an id, a description, a card holder id, the card currency, the card value. A card can be of type Loan, Bank Asset, Personal Asset, Expense, Insurance, Pension, Work, etc. A Personal Asset can be a Car, a Real Estate (House, Apartment, and Terrain), Jewelry, etc. A Bank Asset can be a Deposit Account, a Savings Account, a Credit Card Account, etc. Insurance can be Life Insurance, House Insurance, Car Insurance, etc. A Pension can be a State Pension, or Private Pension. Work can be of type Employment or Freelance. A Loan can be of type Interest Loan or Mortgage. A Mortgage can contain linked to it one or more financial cards, for example one or more houses, terrains, a life insurance, etc.

The application has three types of users: client, financial expert and administrator. A client is able to register in the application, complete his financial data (financial cards) in the platform and see his own financial predictions. He can also save his financial predictions as XML and JSON files. The client can update his own profile and can make a request to the Figlo administrator to delete his account. An expert can create/read/update/delete financial cards in the platform and offer support for clients for their financial predictions. An administrator can do CRUD operations on experts and can delete clients at request.

Requirements:

1. (1P) Identify key actors and use cases. Create the corresponding UML use case diagram.
2. (2P) Create the conceptual architecture of the system. Choose an architectural pattern studied at course. Explain in writing how the system is organized on the chosen pattern.
3. (1p) Create the sequence diagram for deleting a client.
4. (3P) Create the UML class diagram that must contain the important classes with their relevant attributes and methods. Choose and use at least two of the following design patterns that fit the requirements of the Figlo platform: Observable, Composite, Proxy, Command, Strategy, Factory Method, Proxy, Abstract Factory.
5. (1P) Create the database diagram.
6. (2P) Write Java / C# code to illustrate the application of the design patterns, previously chosen.

3. Given the following domain model how would you design the relational data model. Provide a diagram and briefly justify the approach you used.



4. Given the specification below identify suitable DP and provide a sketch of a class diagram. Briefly justify your design.

You need to design a system to allow clients that want to buy a car to configure their options for the desired car. An option is a single feature that buyers can add to a vehicle, usually at an extra cost. However, groups of options bundled together and offered at a single price.

Software Design Exam 2022 – 2

1. Analyze the code below considering the design principles you know. Provide a corrected solution if the case. Assume that the surrounding code is correct.

```

public String hashPassword(String password, HashingType hashingType)
{
    if (HashingType.BASE64.equals(hashingType))
    {
        hashedPassword="hashed with Base64";
    }
    else if (HashingType.MD5.equals(hashingType))
    {
        hashedPassword="hashed with MD5";
    }
    return hashedPassword;
}
  
```

2. Considering the following specification, identify some volatilities based on which a service oriented layered architecture can be developed. Sketch the main components of your architecture. A car assistance system collects data from various input devices (speedometer, cameras, rain sensor etc.) and generates a reaction. The reaction can be either a warning sent to the dashboard or an action such as braking, starting the windshield wipers. The system also learns behavioral patterns and signals outlier behavior.

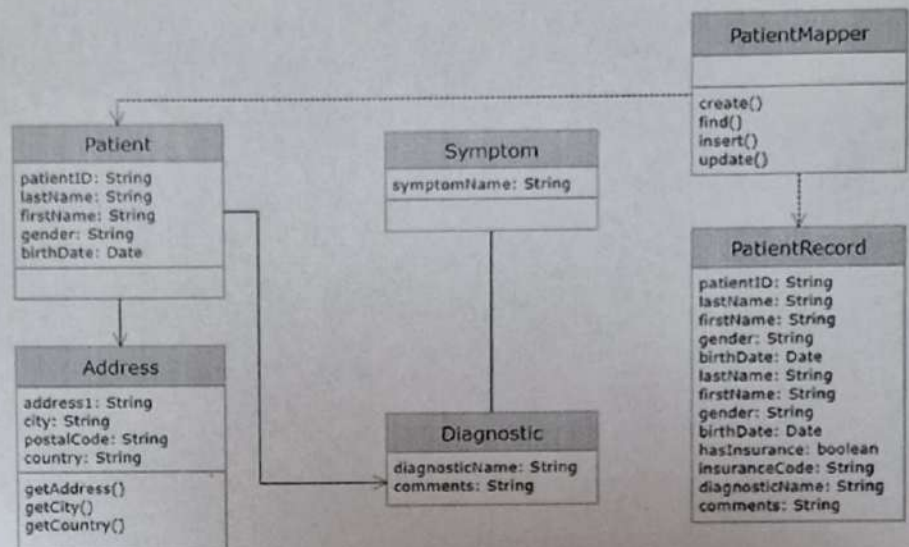
1. Consider an elderly health monitoring system that collects data from various input devices (localization sensors, physiological sensors, speech, etc), stores the data, processes it and generates a reaction. The system learns behavioral patterns of the elderly person. The reaction can be either an alert sent to various systems in the case of a behavioral anomaly or triggering a normal household activity (preheat oven/play music/turn off lights etc..). In case of a behavioral anomaly, the alert needs to be sent in near real-time.

- a) Specify one functional and one non-functional requirement.
- b) Provide a sketch of your architecture. Briefly explain what Architectural Pattern/Style did you use and justify your choice.

2. Consider the following diagram

- a) What patterns are instantiated within the diagram?
- b) What is the corresponding database structure?

3. You have implemented a very complex component that takes some input, does some calculations and saves the results into a text file. You want to sell the component, but first you want to restrict the access to the results by saving the contents in the result text file using some encryption algorithm. a) Suppose you want to be able to use different encryption algorithms. What design pattern(s) can be used here to vary the encryption algorithm easily? b) Sketch the UML diagram of your solution.



NDREEA SAND
MARIA STEKELY
UMB RAZVAN

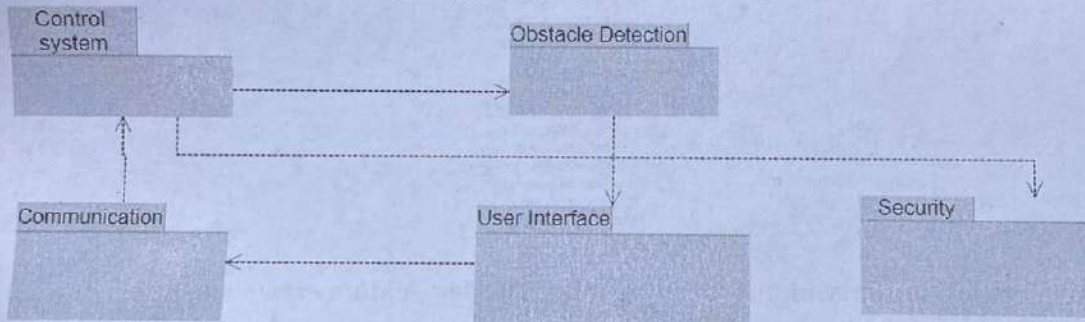
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On the right upper corner please write your FULL NAME, GROUP, Year of passing the lab, Teaching Assistant name.

On the left upper corner write the names of your neighbors (front, back, left and right).

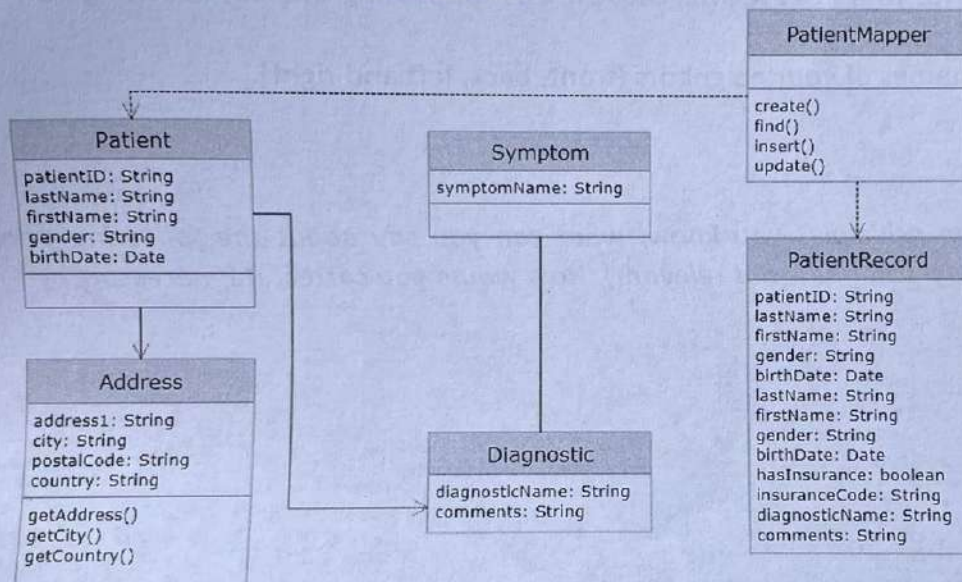
Exam 1

1. Considering the package design principles you know, what can you say about the following package design? (Note. The names of the packages is not relevant). How would you correct it if necessary?



2. Consider a simple elderly health monitoring system that collects data from various input devices (localization sensors, physiological sensors, speech, etc) and generates a reaction. The system learns behavioral patterns. The reaction can be either an alert sent to various systems in the case of a behavioral anomaly or triggering a normal household activity (preheat oven/play music/turn off lights etc..). **Identify some volatilities based on which a service oriented layered architecture can be developed.**

3. Considering the following class design describe the responsibilities of the involved classes. What patterns are instantiated within the diagram?



4. Consider an image recognition system with the following tasks: filtering, feature extraction, labeling, clustering, scene reconstruction; assume that there is no a priori known fixed order in which these have to be called; also, the system should be easy to update with new algorithms.

- i. Which architectural style is most appropriate for the given system? Briefly justify considering the alternatives.
- ii. Give a sketch of sample architecture for the given system. Explain the interaction between the parts of the architecture.

On the right upper corner of your answers paper, please write your FULL NAME, GROUP, Year of passing the lab, Teaching Assistant name.

front

On the left upper corner write the names of your neighbors (left -| - right).

Back

1. *Comment the code snippet below considering the SOLID principles. Provide a better alternative if the case.*

```
public class Vehicle {  
    int velocity;  
    void brake(){velocity--;}  
    Object recognizeRoadSign(){throws UnsupportedOperationException;}}
```

```
public class SelfDrivenCar extends Vehicle {  
    Object recognizeRoadSign () {  
        //classifies Object  
        return recognizedObject;}}
```

2. A JobAgency allows for Seekers to enter their CV containing contact information, education level and skills and Companies to publish job opportunities. When a job opportunity is entered, the agency notifies all the potential interested seekers. Seekers can search themselves for jobs and apply for desired jobs.

Sketch a class design (include a diagram) for this problem that includes the appropriate DP(s).

3. "An online platform stores financial data of customers represented as financial cards. Certain types of cards can have links to other financial cards. A card contains at least: an id, a description, a card holder id, the card currency, the card value. A card can be of type Loan, Bank Asset, Personal Asset, Expense, Insurance, Pension, Work. A Personal Asset can be a Car, a Real Estate, Jewelry. A Bank Asset can be a Deposit Account, a Savings Account, a Credit Card Account. Insurance can be Life Insurance, House Insurance, Car Insurance. A Pension can be a State Pension, or Private Pension. Work can be of type Employment or Freelance. A Loan can be of type Interest Loan or Mortgage. A Mortgage can contain linked to it one or more financial cards."

Sketch a conceptual design (include a diagram) considering the domain driven approach. Specify the types of your components (i.e. Entities, Value Objects, Aggregates etc)

4. A Smart Home System (SHS) keeps track of temperature, electricity and water consumption. The system stores the collected data. A web app allows for querying current values or historical information on a weekly and monthly basis. Additional statistics should be considered to be added. It also notifies users if outlier values are recorded.

i. Which architectural style is most appropriate for the given system? Briefly justify considering the alternatives.

ii. Give a sketch of sample architecture for the given system. Explain the interaction between the parts of the architecture.

Problema

RestoOnline este o platformă de food ordering. Ea agregă mai multe restaurante care își publică fiecare meniul disponibil. Un item de meniu este reprezentat prin titlu, descriere, fotografie, un indicator de disponibilitate și un pret. Sistemul realizează automat calculul pretului comenzii depuse de un client și al timpului în care comanda va fi gata. O comanda este compusă din mai mulți itemi, fiecare cu o cantitate atasată. Fiecare item are un timp de execuție/preparare. Fiecare restaurant are pret de execuție/preparare per minut și un adaos comercial. Execuția unei comenzi se face secvențial, prin prepararea fiecărui item de meniu unul după celălalt. Datele sistemului sunt persistate într-o bază de date. Platforma oferă două tipuri de utilizatori autentificați. **Clienții** pot vizualiza lista de restaurante și selecta restaurantul din care doresc să facă comanda. Ei pot deschide o comanda și să o configureze. Odată ce structura comenzii este definitivată, clientul poate opta pentru checkout, când sistemul îi va afișa pretul total. În urma confirmării comenzii de către client, sistemul va trimite automat un email de notificare care conține datele comenzii și ora estimată la care comanda poate fi ridicată. Clientul poate vizualiza istoricul de comenzi și starea lor. **Angajații** restaurantului pot vizualiza starea comenzilor depuse la restaurantul lor. Ei pot edita detaliile restaurantelor (nume, adresă, profilul culinar, orar, preturi) și meniul restaurantelor, prin adăugarea sau editarea itemilor de meniu. Odată ce o comanda a fost ridicată, angajații o marchează ca fiind finalizată. Angajații pot genera rapoarte cu itemii de meniu cei mai vânduți pe o perioadă anume în format XML/TXT.

1. (1 punct) Să se identifice actorii și să se realizeze diagrama cazurilor de utilizare pentru sistem. Să se realizeze o descriere a cazului de utilizare: "Editare item meniu".
2. (3 puncte) Să se realizeze arhitectura conceptuală a sistemului și diagrama de clase, ținând cont de următoarele cerințe: Se va utiliza un design pattern architectural studiat la curs, exceptând Layers; Se va utiliza design pattern-ul Factory pentru a trata unitar în sistem exportul rapoarte; Se va utiliza design pattern-ul Command pentru a trata calculul de sistem pentru pretul comenzii și al timpului de preparare a comenzii.
3. (1 punct) Realizați diagrama de secvență pt. cazul de utilizare "Afișare comenzi finalizate".
4. (1 punct) Realizați o diagramă a bazei de date pentru aplicația cerută.
5. (3 puncte) Cod (utilizați orice limbaj de programare doriți):
 - a. (1.5 puncte) Să se evalueze folosirea design patternului Factory pentru generarea de rapoarte (nu sunt necesare detalii legate de funcțiile de export în xml/text, trebuie doar evidențiată folosirea design patternului).
 - b. (1.5 puncte) Să se scrie partea de cod corespunzătoare cazului de utilizare "Afișare lista de restaurante" (cuprinzând interacțiunea cu baza de date, în orice modalitate studiată).

Problem

An association of 'sculpture enthusiasts' wants the implementation of an application that makes easier the organizing of new events within their organizations and also sales of sculptures.

The application will have three types of users: guest, normal user and 'event planner'.

A guest will only be able to view scheduled events and offers, available for sculptures.

In addition, a normal user will be able to register to any event, to post new offers and buy sculptures posted by other users. Each sculpture offer must contain at least a price, an author, a description and one or more pictures. Also he can create csv and pdf reports containing the sculptures that he wants to buy.

An 'event planner', in addition to a normal user, will be able to create events. An event can be a meeting for enthusiasts, exhibition or auction and will contain minimally a description, the date when it will occur, the estimated price for an entrance and the minimum number of participants. They may be modified at any time by the 'event planner' that created the event, and for each change, all users who have registered for the event will be notified.

Requirements:

1. (1P) Identify key actors and use cases. Create the corresponding UML use case diagram.
2. (2P) Create the conceptual architecture of the system. Choose an architectural pattern studied at course. Explain in writing how the system is organized on the chosen pattern.
3. (1p) Create the database diagram of the system.
4. (2P) Create the UML class diagram that must contain the important classes with their relevant attributes and methods. Choose and use at least two of the following design patterns: Factory Method, Observable, Composite, Abstract Factory, Proxy, Command, Strategy.
5. (1P) Create the deployment diagram of the system.
6. (2P) Write Java / C # code to illustrate the application of the design patterns, previously chosen.

Software Design 2017

1. Consider an operating system with basic services such as interrupt and exception handling, thread scheduling and dispatching; resource management services such as object management, virtual memory management, I/O management, and security monitor; and, finally, utility programs such as command shell and administration tools. Which architectural style is most appropriate for the given system? Justify your choice. Sketch a sample architecture for the given system. Briefly explain the interaction between the parts of the architecture.
2. You are implementing a file system. The main abstractions in your design would be files and directories. Directories can contain zero or more files or directories. You want to treat directories and files in a uniform way, e.g., both will have name and will provide operations to stream content in and out, and to list children. i) What design pattern could be used to achieve this design? ii) Please explain your design by giving a class diagram.
3. What are the common and different characteristics of the Bridge and the Strategy design patterns? Give examples for applying each of them.
4. Consider an e-mail system containing the classes WebMailSystem, EMail, User, Folder, Attachment and MailBox. Draw a conceptual UML class diagram for the classes and their semantic relationships and cardinalities. Regarding the following functionalities:
 - create a new, empty email
 - move an email to the Trash system folder
 - find emails that contain an attachment with a given name
 - save an attachment to a file
 - register a new user
 - lock a user accountJustify textually based on the principles you know which of the classes is responsible for each function and add the methods to the classes.